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A friendly introduction to mathematical machine learning research

By *Brandon Yea*

Abstract. In this paper, we provide a walk-through into some of the important machine learning research results with a focus on their mathematical aspects. Specifically, we discuss Deep Neural Networks (DNN) and Generative Adversarial Networks (GAN). For DNN, theorems are discussed proving its capability of approximating an arbitrary functional relationship between input and output where mathematical tools from measure theory and functional analysis are heavily used. For GAN, a unifying view of its numerous algorithmic variants is discussed where the concepts from measure theory are again leveraged such as push-forward measure, absolute continuity of measure, and the metrics quantifying the difference between measures. As these examples represent the most fundamental and popular building blocks of modern machine learning algorithms, this paper will provide a useful insight into the connections between advanced mathematical ideas and the burgeoning area of machine learning.

1 Introduction

In this expository paper, we provide a walk-through into some of the classic machine learning research papers. The purpose is to provide the reader with an introduction to a landscape of machine-learning research where the arsenal of measure-theoretic functional analysis plays a vital role in understanding and developing the underpinning theories that lead and support algorithmic progresses in the field.

The construction of the paper is as follows: In Section 2, we briefly describe trends and major approaches in ML algorithm research. We then move on to specific examples where many concepts and tools from mathematical analysis are leveraged. Specifically, in Section 3, we first introduce some relevant results from measure theory and functional analysis and discuss examples of analyzing the Neural Network as a universal approximator. A similar effort is made in Section 4 where we discuss the Generative Model which is arguably one of the most popular driving forces behind the recent rise of machine-learning technology. We conclude the paper in Section 5.

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2 Background

2.1 Brief Introduction to machine learning

The field of artificial intelligence was born in the 1950s and entails the question of whether computers could be made to think as humans. Representative approaches include symbolic AI which pursued human-level intelligence based on the premise that it could be achieved by having a sufficient number of explicit rules for manipulating knowledge. However, beyond certain well-defined and logical problems, it was found to be difficult to come up with rules for solving complex and fuzzy problems such as language translation and image recognition.

Machine learning systems, on the other hand, learn the rules for a given task rather than just performing the task according to the given rules. Therefore a machine-learning system is trained based on the input data and the expected answers. More specifically, it is trained to learn a useful representation of the relationship between the input and the output - the expected answers.

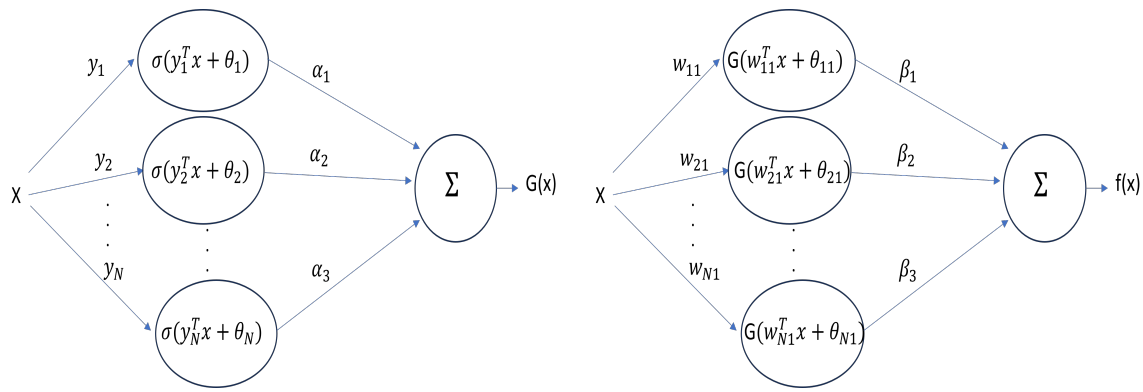
2.2 Supervised vs. unsupervised learning

If the representation is learned with both the input and the desired output data, the machine learning algorithm belongs to the category of *supervised* learning. Most traditional machine learning algorithms are in this category. However, very often, there are cases where the desired output cannot be obtained or specified for each input. In such scenarios, the algorithm can be designed to train itself in such a way that a certain goal hence its associated metric is optimized without relying on a set of target outputs. These classes of machine learning algorithms belong to *unsupervised* learning.

In Section 3, we will study the mathematical analysis of the representation obtained based on the architecture called the **DNN (Deep Neural-Network)**, which belongs to the category of supervised learning. In Section 4, we will study the algorithm called **GAN (Generative Adversarial Network)**, which is a form of unsupervised learning, where the aforementioned neural-network is used as its key module and trained without relying on explicitly-provided target output.

3 Approximation capability of deep neural networks

In this section, we study two classic papers [8][9] from neural-network research that entail the approximation capability of neural-networks. The mathematical analyses of this kind often turn out to be critical in setting our expectations about what the neural networks are capable of. They can provide certain guarantees or estimates about its



(a) NN structure in Cybenko[8]

(b) NN structure in Hornik[9]

Figure 1: Single hidden-layer neural networks

outputs assuming the input and output can be described as a function belonging to a certain class. Interestingly, these papers appeared around 1990 and demonstrated similar results with different mathematical approaches and assumptions. The results actually pertain to a neural network with a single hidden-layer (aka shallow network) with infinitely many neuron units. Figure 1 shows examples of such a network used in the subsequent discussions in this paper. However, lots of subsequent works followed that proved similar approximation capability of 'deep' networks which have many layers (see [11] for example). In fact, current popularity of deep learning is based on the use of this deep networks for which a few important progresses were made in terms of its training and the availability of big data as well as fast-computation. Regardless, the mathematical approaches in these two papers have been and will continue to be good examples that merit thorough study for future research in this area.

3.1 Preliminaries

The first paper by Cybenko[8] we will study in Section 3.2 relies on Hahn-Banach Theorem and Riesz Representation Theorem. We will provide these theorems and a few lemmas that would be needed to prove the main results in the paper. We start with some notations and basic concepts first.

Definition 3.1. Some notations of particular sets of interest are given below:

- C^n : the set of continuous functions from R^n to R
- M^n : the set of all Borel-measurable functions from R^n to R
- B^n : the Borel σ-field of R^n

Remark 3.2. Closeness of functions f and g belonging to C^n or M^n is measured by a **metric**, ρ, while closeness of one class of functions to another is described by **denseness**.

Definition 3.3. A subset S of a metric space (X, ρ) is **ρ-dense** in a subset T if for every $\epsilon > 0$ and for every $t \in T$, there is an $s \in S$ such that $\rho(s, t) < \epsilon$. This means an element of S can approximate an element of T to any desired degree of accuracy.

Definition 3.4. Some notions about functionals on L^p space are summarized below:

- **Linear functional** A linear functional on L^p is a mapping G of L^p into R such that

$$G(af + bg) = aG(f) + bG(g)$$

  for all $a,b \in R$ and $f,g \in L^p$.

- **Bounded linear functional** The linear functional G is bounded if there exists a constant M such that

$$|G(f)| \leq M||f||_p$$

  for all $f \in L^p$.

- **Norm of a linear functional** The bound or the norm of a linear functional G is defined to be

$$||G|| = \sup\{|G(f)| : f \in L^p, ||f||_p \leq 1\}.$$

Theorem 3.5. ***Hahn-Banach*** *(see Theorem5.16 of [6]) Suppose* V *is a normed vector space,* U *is a subspace of* V*, and* $\Psi : U \mapsto F$ *is a bounded linear functional. Then* Ψ *can be extended to a bounded linear functional on* V *whose norm equals* $||\Psi||$.

Theorem 3.6. ***Riesz Representation(I),[4]*** *If* $(X, \mathscr{X}, \mu)$ *is a* σ*-finite measure space and* G *is a bounded linear functional on* $L^1(X, \mathscr{X}, \mu)$*, then there exists* $g \in L^\infty(X, \mathscr{X}, \mu)$ *such that*

$$G(f) = \int fg d\mu$$

holds for all $f \in L^1$*. Moreover,* $||G|| = ||g||_\infty$ *and* $g \geq 0$ *if* G *is a positive linear functional.*

Theorem 3.7. ***Riesz Representation(II),[4]*** *If* G *is a bounded positive linear functional on* $C(I_n)$*, which is the Banach space of all continuous functions on* I_n *to* R *with the norm* $||f|| = \sup\{|f(x)| : x \in I_n\}$*, then there exists a measure* μ *defined on the Borel subsets of* R *such that*

$$G(f) = \int_{I_n} f d\mu$$

for all $f \in C(I_n)$*. Moreover, the norm* $||G||$ *of* G *equals* $\mu(I_n)$.

Lemma 3.8. *Let* U *be a linear, non-dense subspace of a normed linear space* X. *Then there exists a bounded linear functional* $\mathscr{L}$ *on* X *such that* $\mathscr{L} \neq 0$ *on* X *and* $\mathscr{L}_{|\mathrm{U}} = 0$.

Proof. This lemma can be proved using Theorem 3.5 (Hahn-Banach Theorem). See Theorem 5.19 of [6]. □

Lemma 3.9. *Let* U *be a linear, non-dense subspace of* $\mathrm{C}(\mathrm{I}_n)$. *Then there is a measure* $\mu \in \mathrm{M}(\mathrm{I}_n)$ *such that*

$$\int_{\mathrm{I}_n} h d\mu = 0 \textit{ for all } h \in \mathrm{U}$$

Proof. If we let $\mathrm{X} \triangleq \mathrm{C}(\mathrm{I}_n)$ in lemma 3.8, there is a bounded linear functional $\mathscr{L} : \mathrm{C}(\mathrm{I}_n) \to \mathrm{R}$ such that $\mathscr{L} \neq 0$ on $\mathrm{C}(\mathrm{I}_n)$ and $\mathscr{L}_{|\mathrm{U}} = 0$. Applying Theorem 3.7 (Riesz representation theorem of linear bounded functionals on $\mathrm{C}(\mathrm{I}_n)$), there is a measure $\mu \in \mathrm{M}(\mathrm{I}_n)$ such that

$$\mathscr{L}(f) = \int_{\mathrm{I}_n} f d\mu \text{ for all } f \in c(\mathrm{I}_n) \tag{1}$$

In particular, for any $h \in \mathrm{U}$, we obtain

$$\mathscr{L}(h) = \int_{\mathrm{I}_n} h d\mu = 0 \tag{2}$$

□

3.2 Approximation by superpositions of a sigmoidal function

We first look at theorems from the paper by Cybenko[8] that tries to identify the classes of functions that can be efficiently realized by artificial neural networks. More specifically, the theorems demonstrate that networks with only one internal layer and an enough number of arbitrary continuous or bounded measurable sigmoidal functions can well approximate arbitrary continuous functions (Theorem 3.15) or functions in the L^1 space (Theorem 3.16), respectively. Figure 1a shows the architecture of the neural-network used in these theorems.

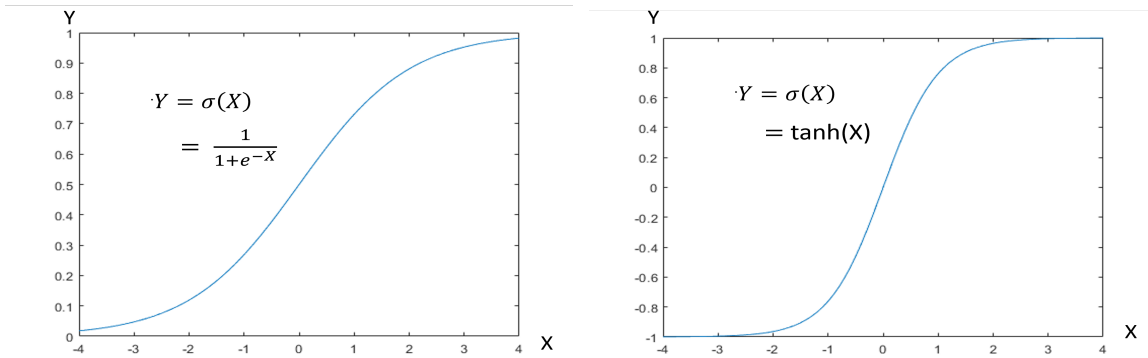
Definition 3.10. Sigmoidal function A function $\sigma : \mathrm{R} \to [0,1]$ is called sigmoidal if

$$\lim_{x \to -\infty} \sigma(x) = 0, \ \lim_{x \to +\infty} \sigma(x) = 1$$

Definition 3.11. Discriminatory function We say that σ is discriminatory if for a measure $\mu \in \mathrm{M}(\mathrm{I}_n)$

$$\int_{\mathrm{I}_n} \sigma(y^{\mathrm{T}} + \theta) d\mu(x) = 0$$

for all $y \in \mathrm{R}^n$ and $\theta \in \mathrm{R}$ implies that $\mu = 0$.



(a) Logistic function (b) Hyperbolic tangent function

Figure 2: Examples of sigmoidal functions

Remark 3.12. Note the sigmoidal function defined above is used as a unit of computation in each layer of neural networks as illustrated in Figure 1a, where we can see $\sigma(\cdot)$ is an "activation" function that takes in a weighted-sum of input values and emits either a binary output based upon a threshold or a continuous-valued output. Popular choices of such a function include the logistics function and the hyperbolic tangent function depicted in Figure 2a. By allowing this unit to be an arbitrary non-linear function including the sigmoidal function, one might expect the approximation capability of the network to be better than that of a linear combination of inputs. In fact, the beauty of the mathematical results we will visit in this regard is that we can guarantee its asymptotic approximation capability assuming a rather broad class of functions subject to the conditions such as being discriminatory.

The following theorem shows the class of discriminatory functions well approximates the continuous function. We leverage this result together with Lemma 3.14 to arrive at Theorem 3.15, then use a similar argument to prove Theorem 3.16.

Theorem 3.13. ***(Cybenco,[8])*** *Let σ be any continuous discriminatory function. Then finite sums of the form*

$$G(x) = \sum_{j=1}^{N} \alpha_j \sigma(y_j^T x + \theta_j) \tag{3}$$

are dense in $C(I_n)$. In other words, given any $f \in C(I_n)$ and $\epsilon > 0$, there is a sum, $G(x)$, of the above form, for which

$$|G(x) - f(x)| \leq \epsilon \quad \textit{for all } x \in I_n.$$

Proof. Let $S \subset C(I_n)$ be the set of functions of the form $G(x)$ as in (3). Clearly S is a linear subspace of $C(I_n)$. Now assume S is not dense in $C(I_n)$. By lemma 3.9, there is a measure

$\mu \in M(I_n)$ such that

$$\int_{I_n} h d\mu = 0 \text{ for all } h \in S$$

,which leads to

$$\sum_{j=1}^{N} \alpha_j \int_{I_n} \sigma(y_j^T x + \theta_j) d\mu = 0$$

By selecting α_j's judiciously, we obtain

$$\int_{I_n} \sigma(y^T + \theta) d\mu = 0 \text{ for all } y \in R^n, \theta \in R$$

Since σ is discriminatory, it requires $\mu = 0$, which is a contradiction. Hence, the subspace S must be dense in $C(I_n)$

□

Lemma 3.14. ***(Cybenco,[8])*** *Any bounded, measurable sigmoidal function,* σ*, is discriminatory for all measures* $\mu \in M(I_n)$*. In particular, any continuous sigmoidal function is discriminatory.*

Proof. Let us consider a bounded measurable sigmoidal function σ such that

$$\int_{I_n} \sigma(w^T x + \theta) d\mu(x) = 0 \tag{4}$$

for all $w \in R^n$ and $\theta \in R$, given a measure $\mu \in M(I_n)$. According to Definition 3.11, we need to show $\mu = 0$ to prove the lemma. Note the continuous sigmoidal function is also bounded (by definition of sigmoidal function) and measurable for measures in $M(I_n)$.

First, if we construct $\sigma_\lambda(x) \triangleq \sigma(\lambda(w^T x + \theta) + \varphi)$, then, by the definition of a sigmoidal, we have

$$\lim_{\lambda \to \infty} \sigma_\lambda(x) = \begin{cases} 1, \text{ for } w^T x + \theta > 0 \\ 0, \text{ for } w^T x + \theta < 0 \\ \sigma(\varphi), \text{ for } w^T x + \theta = 0 \end{cases}$$

Therefore, we have $\sigma_\lambda(x) \to \gamma(x)$ pointwise on R as $\lambda \to \infty$, where a bounded function $\gamma(x)$ is defined as

$$\gamma(x) = \begin{cases} 1, \text{ for } x \in H_{w,\theta} \\ 0, \text{ for } x \in H^-_{w,\theta} \\ \sigma(\varphi), \text{ for } x \in P_{w,\theta} \end{cases}$$

where $\mathrm{H}_{w,\theta} = \{x : w^{\mathrm{T}}x+\theta > 0\}$, $\mathrm{H}^{-}_{w,\theta} = \{x : w^{\mathrm{T}}x+\theta < 0\}$, and $\mathrm{P}_{w,\theta} = \{x : w^{\mathrm{T}}x+\theta = 0\}$. Since $\sigma_\lambda(x)$ is a bounded measurable sequence of λ, we can apply Bounded Convergence Theorem (see Chapter 4 of [2]) to switch the order of the limit and the integral as follows:

$$\begin{aligned}\lim_{\lambda\to\infty}\int_{\mathrm{I}_n}\sigma_\lambda(x)d\mu(x) &= \int_{\mathrm{I}_n}\lim_{\lambda\to\infty}\sigma_\lambda(x)d\mu(x)\\ &= \int_{\mathrm{I}_n}\gamma(x)d\mu(x)\\ &= \int_{\mathrm{H}_{w,\theta}}\gamma(x)d\mu(x)+\int_{\mathrm{H}^{-}_{w,\theta}}\gamma(x)d\mu(x)+\int_{\mathrm{P}_{w,\theta}}\gamma(x)d\mu(x)\\ &= \mu(\mathrm{H}_{w,\theta})+\sigma(\varphi)\mu(\mathrm{P}_{w,\theta})\end{aligned} \tag{5}$$

Since (4) implies $\int_{\mathrm{I}_n}\sigma_\lambda(x)d\mu(x) = \int_{\mathrm{I}_n}\sigma(\lambda w^{\mathrm{T}}x+\lambda\theta+\varphi)d\mu(x) = 0$, we have the left-side of (5) zero, hence obtain

$$\mu(\mathrm{H}_{w,\theta})+\sigma(\varphi)\mu(\mathrm{P}_{w,\theta}) = 0. \tag{6}$$

Furthermore, as (6) is true for $\varphi\to\infty$ and $\varphi\to-\infty$, by the properties of σ, we have $\mu(\mathrm{H}_{w,\theta})+\mu(\mathrm{P}_{w,\theta}) = 0$ and $\mu(\mathrm{H}_{w,\theta}) = 0$, respectively, to obtain

$$\mu(\mathrm{H}_{w,\theta}) = \mu(\mathrm{P}_{w,\theta}) = 0 \tag{7}$$

for all $w\in\mathrm{R}^n$ and $\theta\in\mathrm{R}$.

Next, we will consider the linear functional $\mathrm{F} : \mathrm{L}^\infty(\mathrm{R})\to\mathrm{R}$ defined by

$$\mathrm{F}(h) = \int_{\mathrm{I}_n} h(w^{\mathrm{T}}x)d\mu(x),$$

where $w\in\mathrm{R}^n$ and h is a bounded measurable function, and show $\mu = 0$ if $\mathrm{F}(h) = 0$ for any $h\in\mathrm{L}^\infty(\mathrm{R})$.

First, note F is a bounded functional as it satisfies the following inequality:

$$|\mathrm{F}(h)| = |\int_{\mathrm{I}_n} h(w^{\mathrm{T}}x)d\mu(x)| \le ||h||_\infty|\int_{\mathrm{I}_n} d\mu(x)| = ||h||_\infty|\mu(x)| < \infty$$

as μ is a finite measure. By letting h be the indicator function of the interval $[\theta,\infty)$ as denoted by $h = 1_{[\theta,\infty)}$, we have

$$\mathrm{F}(h) = \int_{\mathrm{I}_n} h(w^{\mathrm{T}}x)d\mu(x) = \int_{\{w^{\mathrm{T}}x\ge\theta\}} d\mu(x) = \mu(\mathrm{P}_{w,-\theta})+\mu(\mathrm{H}_{w,-\theta}) = 0.$$

by (7). Similarly, if h is the indicator function of the open interval (θ,∞) as denoted by $h = 1_{(\theta,\infty)}$, we obtain

$$\mathrm{F}(h) = \int_{\mathrm{I}_n} h(w^{\mathrm{T}}x)d\mu(x) = \int_{\{w^{\mathrm{T}}x>\theta\}} d\mu(x) = \mu(\mathrm{H}_{w,-\theta}) = 0$$

Since the indicator of any interval can be written as a linear combination of the above two types of indicator functions, we have $F(h) = 0$ for any simple function h as a simple function is again a linear combination of indicator functions. Moreover, we note simple functions are dense in $L^\infty(R)$ (see Section 7.4 of [2]). In other words, for any fixed $h \in L^\infty(R)$, there is a sequence of simple functions $\{S_n\}$ such that $S_n \to h$. It follows that, for any $h \in L^\infty(R)$,

$$F(h) = F(\lim_{n\to\infty} S_n) = \lim_{n\to\infty} F(S_n) = 0 \tag{8}$$

, where we used the fact F is bounded and continuous.

Finally, we obtain the Fourier Transform of the measure μ as

$$\begin{aligned}\hat{\mu} &= \int_{I_n} \exp^{iw^T x} d\mu(x) = \int_{I_n} \cos(w^T x) d\mu(x) + i \int_{I_n} \sin(w^T x) d\mu(x) \\ &= F(\cos(\cdot)) + iF(\sin(\cdot)) = 0\end{aligned}$$

for all $w \in R^n$ using (8) and the fact that sin() and cos() belong to $L^\infty(R)$, which leads to $\mu = 0$ since the inverse Fourier Transform of zero is zero.

Therefore, we proved $\mu = 0$ when $F(h) = 0$ for any $h \in L^\infty(R)$, which implies, in particular, $\mu = 0$ when (4) is true. □

Theorem 3.15. ***(Cybenco,[8])*** *Let σ be any continuous sigmoidal function. Then finite sums of the form*

$$G(x) = \sum_{j=1}^{N} \alpha_j \sigma(y_j^T x + \theta_j) \tag{9}$$

are dense in $C(I_n)$. In other words, given any $f \in C(I_n)$ and $\epsilon > 0$, there is a sum, $G(x)$, of the above form, for which

$$|G(x) - f(x)| < \epsilon \text{ for all } x \in I_n$$

Proof. Combine Theorem 3.13 and Lemma 3.14, noting that continuous sigmoids satisfy the conditions of that lemma. □

Theorem 3.16. ***(Cybenco,[8])***

Let σ be bounded measurable sigmoidal function. Then finite sums of the form

$$G(x) = \sum_{j=1}^{N} \alpha_j \sigma(y_j^T x + \theta_j) \tag{10}$$

are dense in $L^1(I_n)$. In other words, given any $f \in L^1(I_n)$ and $\epsilon > 0$, there is a sum $G(x)$ of the above form for which

$$||G - f||_{L^1} = \int_{I_n} |G(x) - f(x)| dx < \epsilon$$

Proof. Assume a linear subspace of $L^1(I_n)$

$$U = \{G : G(x) = \sum_{j=1}^{N} \alpha_j \sigma(w_j^T x + \theta_j)\}$$

is not dense in $L^1(I_n)$. By lemma 3.8, there exists a bounded linear functional $F : L^1(I_n) \to R$ such that $F \neq 0$ on $L^1(I_n)$ and $F_{|U} = 0$. Noting that $L^\infty(I_n)$ is the dual of $L^1(I_n)$, Theorem 3.6 guarantees the existence of $g \in L^\infty(I_n)$ such that

$$F(f) = \int_{I_n} f(x)g(x)dx \text{ with } ||F|| = ||g||_1 \neq 0.$$

Now, writing the condition $F_{|U} = 0$ as

$$\int_{I_n} G(x)g(x)dx = 0 \text{ for all } G \in U,$$

we have $\int_{I_n} \sigma(w^T x + \theta)g(x)dx = 0$.

Since σ is bounded and measurable, by lemma 3.14, it is discriminatory for the measure $\mu(x) \triangleq g(x)dx$ which belongs to $M(I_n)$ given any $g(x) \in L^\infty(I_n)$. Therefore we require $g(x) = 0$, which contradicts $||F|| \neq 0$. It follows that U is dense in $L^1(I_n)$. □

3.3 Universal approximation with a single hidden layer

Next, we look at a paper by Hornik [9] that shows a single-layer neural network with continuous 'neurons' can again well approximate any continuous (Theorem 3.24) or Borel-measurable (Theorem 3.29) functions. Similar results as in Section 3.2 are obtained, but it covers a broader class of functions as approximator. Also the center piece of the mathematical machinery here is the Stone-Weierstrass Theorem, which is provided below without proof.

Theorem 3.17. ***Stone-Weierstrass theorem [5]*** *Let* A *be an algebra of real continuous functions on a compact set* K*. If* A *separates points on* K *and if* A *vanishes at no point of* K*, then the uniform closure* B *of* A *consists of all real continuous functions on* K *(i.e.,* A *is* ρ_K*-dense in the space of real continuous functions on* K*).*

Remark 3.18. Notes on the terms appearing in the statements of the above theorem

- A family A of real functions defined on a set E is an **algebra** if A is closed under addition,multiplication, and scalar multiplication.
- If A has the property that $f \in A$ whenever $f_n \in A$ and $f_n \to f$ uniformly on E, then A is said to be **uniformly closed**.

- Let B be the set of all functions which are limits of uniformly convergent sequences of members of A. Then B is called the **uniform closure** of A.
- A family A **separates** points on E if for every x, y in E such that $x \neq y$, there exists a function f in A such that $f(x) \neq f(y)$. Also a family A **vanishes at no point** of E if for each x in E, there exists f in A such that $f(x) \neq 0$.

We begin by defining the class of neural network representation we will use.

Definition 3.19. For any measurable function $\mathrm{G}(\cdot)$ mapping R to R and $r \in \mathrm{N}$, let $\boldsymbol{\Sigma\Pi^r(\mathrm{G})}$ be the class of functions

$$\{f : \mathrm{R}^r \to \mathrm{R} : f(x) = \sum_{j=1}^{p} \beta_j \prod_{k=1}^{q_j} \mathrm{G}(w_{jk}^{\mathrm{T}} x + \theta_{jk})\}, \tag{11}$$

where $w_{jk} \in \mathrm{R}^r$, $\beta_j, \theta_{jk} \in \mathrm{R}$, $x \in \mathrm{R}^r$, and $p, q_j \in \mathrm{N}$.

Note that the class $\Sigma\Pi^n(\mathrm{G})$ belongs to M^n for any Borel measurable G. When G is continuous, they all belong to C^n. Figure 1b illustrates the architecture of $\boldsymbol{\Sigma\Pi^r(\mathrm{G})}$ when $p = \mathrm{N}$ and $q_j = 1$.

The first result (Theorem 3.24) concerns approximating functions in C^n, which will come after a few definitions.

Definition 3.20. A subset S of C^n is said to be **uniformly dense on compacta** in C^n if for every compact subset $\mathrm{K} \subset \mathrm{R}^n$, S is ρ_k-dense in C^n, where for $f, g \in \mathrm{C}^n$, $\rho_k(f, g) \equiv \sup_{x \in \mathrm{K}} |f(x) - g(x)|$.

Definition 3.21. Let μ be a probability measure on the measurable space $(\mathrm{R}^n, \mathrm{B}(\mathrm{R}^n))$. We say two measurable functions f and g are **μ-equivalent** if $\mu\{x \in \mathrm{R}^n : f(x) \neq g(x)\} = 0$, or equivalently $\mu\{x \in \mathrm{R}^n : f(x) = g(x)\} = 1$.

Remark 3.22. The metric on classes of μ-equivalent functions relevant for our results is given by the next definition. Note that two functions are close in this metric if and only if there is only a small probability that they differ significantly. If f and g are μ-equivalent, $\rho_\mu(f, g)$ equals zero.

Definition 3.23. Given a probability measure μ on $(\mathrm{R}^r, \mathrm{B}^r)$, define the metric $\boldsymbol{\rho_\mu}$ from $\mathrm{M}^r \times \mathrm{M}^r$ to R^+ by $\rho_\mu(f, g) = \inf\{\epsilon > 0 : \mu\{x : |f(x) - g(x)| > \epsilon\} < \epsilon\}$.

Theorem 3.24. ***(Hornik,[9])*** *Let* G *be any continuous nonconstant function from* R *to* R. *Then* $\boldsymbol{\Sigma\Pi^r(\mathrm{G})}$ *as defined in (11) is uniformly dense in* C^r.

Proof. We will apply the Stone-Weierstrass Theorem by first noting that, for any G, $\Sigma\Pi^r(\mathrm{G})$ is obviously an algebra on an arbitrary compact set $\mathrm{K} \subset \mathrm{R}^r$.

Now, if we let $x, y \in \mathrm{K}$, $x \neq y$, then we need to show there is $f \in \Sigma\Pi^r$ such that $f(x) \neq f(y)$. Since G is nonconstant, there are $a, b \in \mathrm{R}, a \neq b$ with $\mathrm{G}(a) \neq \mathrm{G}(b)$. Pick x and y from the hyperplanes $\{w^{\mathrm{T}}x + \theta = a\}$ and $\{w^{\mathrm{T}}x + \theta = b\}$, respectively. Then $f(u) = \mathrm{G}(w^{\mathrm{T}}u + \theta)$ separates x and y. We can see this by noting that $f(x) = \mathrm{G}(w^{\mathrm{T}}x + \theta) = \mathrm{G}(a)$ and $f(y) = \mathrm{G}(w^{\mathrm{T}}y + \theta) = \mathrm{G}(b) \neq \mathrm{G}(a)$.

Second, let θ satisfy $\mathrm{G}(\theta) \neq 0$ and choose a vector $w^{\mathrm{T}} = (0, \ldots, 0) \in \mathrm{R}^r$. Then $f(x) \equiv \mathrm{G}(w^{\mathrm{T}}x + \theta) = \mathrm{G}(\theta) \neq 0$ is a non-zero constant for all $x \in \mathrm{K}$. This ensures that $\Sigma\Pi^r(\mathrm{G})$ vanishes at no point of K. The Stone-Weierstrass Theorem thus implies that $\Sigma\Pi^r(\mathrm{G})$ is ρ_{K}-dense in the space of real continuous functions on K. Because K is arbitrary, the result follows. □

Next, we present Lemmas 3.25, 3.27, and 3.28 first to prove Theorem 3.29 which extends the result of Theorem 3.24 to the set M^r of Borel-measurable functions.

Lemma 3.25. *All of the following are equivalent.*

(a) $\rho_u(f_n, f) \to 0$.

(b) For every $\epsilon > 0$, $\mu\{x : |f_n(x) - f(x)| > \epsilon\} \to 0$.

(c) $\int \min\{|f_n(x) - f(x)|, 1\}\mu(dx) \to 0$.

Proof. (a) → (b):
Assume (a) holds. Then by definition, for any $\epsilon > 0$ such that $\mu(|f_j - f| > \epsilon) < \epsilon$, we have $\rho_\mu(f_j, f) \leq \epsilon$. Since $\rho_\mu(f_j, f) \to 0$, we may choose $\epsilon \triangleq \frac{1}{n_j} \to 0$ such that

$$\mu(|f_j - f| > \frac{1}{n_j}) < \frac{1}{n_j} \tag{12}$$

If we assume (b) does not hold, there exists an $\epsilon_0 > 0$ such that $\mu\{|f_j - f| > \epsilon\} > \epsilon_0$. This means $\mu\{|f_j - f| > \frac{1}{n_j}\} > \epsilon_0$ when $\epsilon \triangleq \frac{1}{n_j}$ even as $j \to \infty$, which is contradictory to (12).
(b) → (a): obvious.
(b) $\iff$ (c):
Let $\epsilon \in (0, 1)$ be fixed and note

$$\epsilon 1_{(\epsilon, +\infty)}(u) \leq \min(u, 1) \leq \epsilon + 1_{(\epsilon, +\infty)}(u) \text{ for } x \geq 0 \tag{13}$$

If we let $u \triangleq |f_j(x) - f(x)|$ in (13) and note $\int_{\mathrm{R}^n} 1_{|f_j(x) - f(x)| > \epsilon} d\mu(x) = \mu\{x : |f_j(x) - f(x) > \epsilon\}$, we have the following two equations:

$$\epsilon\mu\{x : |f_j(x) - f(x)| > \epsilon\} \leq \int_{\mathrm{R}^n} \min(|f_j(x) - f(x)|, 1) d\mu(x) \tag{14}$$

$$\epsilon + \mu\{x : |f_j(x) - f(x)| > \epsilon\} \geq \int_{\mathrm{R}^n} \min(|f_j(x) - f(x)|, 1)\, d\mu(x) \tag{15}$$

If (c) is true, (14) implies (b). On the other hand, if (b) is true, (15) implies

$$\epsilon \geq \lim_{j\to\infty} \int_{\mathrm{R}^n} \min(|f_j(x) - f(x)|, 1)\, d\mu(x),$$

which leads to (c). □

Definition 3.26. A sequence of functions $\{f_n\}$ converges to a function f uniformly on compacta if for all compact $\mathrm{K} \subset \mathrm{R}^n$, $\rho_{\mathrm{K}}(f_n, f) \to 0$ as $n \to \infty$.

Lemma 3.27. ***(Hornik,[9])*** *If $\{f_n\}$ is a sequence of functions in M^r that converges uniformly on compacta to the function f then $\rho_\mu(f_n, f) \to 0$.*

Proof. Let us pick an $\epsilon > 0$, then, by Lemma 3.25, it is sufficient to show part (c), which says we can find $\mathrm{N} \in \mathcal{N}$ such that for all $j \geq \mathrm{N}$ we have $\int \min\{|f_j(x) - f(x)|, 1\}\mu(dx) < \epsilon$.

Without loss of generality, let us suppose $\mu(\mathrm{R}^n) = 1$. Since R^n is a locally compact metric space, it can be represented as the union of a sequence $\{\mathrm{K}_j\}$ of compact sets (e.g., a sequence of closed balls with increasing diameters) and μ is a regular measure such that $\mu(\mathrm{R}^n) = \sup\{\mu(\mathrm{K}) : \mathrm{K} \subset \mathrm{R}^n \text{ and K is compact}\}$(see [7],chap 7). In other words, there is a compact subset K of R^n with $\mu(\mathrm{K}) > 1 - \epsilon/2$ so that $\mu(\mathrm{R}^n \setminus \mathrm{K}) = \mu(\mathrm{R}^n) - \mu(\mathrm{K}) < \epsilon/2$.

Next, note the assumption that says the sequence f_n converges uniformly on compacta to f, we have $\sup_{x\in\mathrm{K}} |f_j(x) - f(x)| \to 0$ as $j \to \infty$, which is equivalent to say we can find $\mathrm{N} \geq 0$ such that $\sup_{x\in\mathrm{K}} |f_j(x) - f(x)| < \epsilon/2$ for all $j \geq \mathrm{N}$.

Consequently, we have the following inequality:

$$\begin{aligned}
&\int_{\mathrm{R}^n} \min(|f_j(x) - f(x)|, 1)\, d\mu(x) \\
&= \int_{\mathrm{R}^n\setminus\mathrm{K}} \min(|f_j(x) - f(x)|, 1)\, d\mu(x) + \int_{\mathrm{K}} \min(|f_j(x) - f(x)|, 1)\, d\mu(x) \\
&\leq \int_{\mathrm{R}^n\setminus\mathrm{K}} 1 \cdot d\mu(x) + \int_{\mathrm{K}} \min(\sup_{x\in\mathrm{K}} |f_j(x) - f(x)|, 1)\, d\mu(x) \\
&= \mu(\mathrm{R}^n \setminus \mathrm{K}) + \int_{\mathrm{K}} \min(\sup_{x\in\mathrm{K}} |f_j(x) - f(x)|, 1)\, d\mu(x) \\
&< \epsilon/2 + \int_{\mathrm{K}} \sup_{x\in\mathrm{K}} |f_j(x) - f(x)|\, d\mu(x) < \epsilon/2 + \epsilon/2 \cdot \mu(\mathrm{K}) < \epsilon/2 + \epsilon/2 = \epsilon \text{ for all } j \geq \mathrm{N}.
\end{aligned}$$

, which is part (c) of Lemma 3.25. □

Lemma 3.28. ***(Hornik,[9])*** *For any finite measure μ, C^r is ρ_μ-dense in M^r.*

Proof. Note we must find $g \in C^r$ such that $\rho_\mu(f,g) < \epsilon$. for an arbitrary $f \in M^r$ and $\epsilon > 0$.

First, let us consider $A_n \triangleq \{x : |f(x)| > n\}$, then since $A_{n+1} \subset A_n$ and $A_n \in B(R^n)$, $\{A_n\}$ is a decreasing sequence of measurable sets with $\lim A_n = \bigcap_n A_n = \{x : |f(x)| = \infty\}$.

From the sequential continuity and the fact f is finite a.e., we get $\mu(A_n) \to 0$ as $n \to \infty$. Therefore we can choose M large enough such that $\mu\{|f| > M\} < \epsilon/2$ and obtain $\int \min(|f \cdot 1_{\{|f|<M\}} - f|, 1) d\mu = \int_{|f|\geq M} \min(|f|, 1) d\mu = \int_{|f|\geq M} d\mu = \mu\{|f| > M\} < \epsilon/2$.

On the other hand, it can be shown (see [3] Theorems 55.C and D. p. 241-242) there is a continuous g such that $\int |f \cdot 1_{\{|f|<M\}} - g| d\mu < \epsilon/2$.

By Triangle inequality, we have the following:

$$\begin{aligned} \int \min(|f-g|,1) d\mu &\leq \int \min(|f - f \cdot 1_{\{|f|<M\}}|, 1) d\mu + \int \min(|f \cdot 1_{\{|f|<M\}} - g|, 1) d\mu \\ &< \epsilon/2 + \int |f \cdot 1_{\{|f|<M\}} - g| d\mu \\ &< \epsilon/2 + \epsilon/2 = \epsilon \end{aligned} \tag{16}$$

By Lemma 3.25, we have $\rho_u(g,f) \to 0$, which shows C^r is ρ_μ-dense in M^r since $g \in C^r$ and $f \in M^r$. □

Now we state the result that extends the first one (Theorem 3.24) to approximating functions in M^n

Theorem 3.29. ***(Hornik,[9])***
For every continuous nonconstant function G*, every* r*, and every probability measure* μ *on* (R^r, B^r)*,* $\Sigma\Pi^r(G)$ *is* ρ_μ*-dense in* M^r*.*

Proof. Given any continuous nonconstant function, it follows from Theorem 3.24 and Lemma 3.27 that $\Sigma\Pi^r(G)$ is ρ_μ-dense in C^r. Because C^r is ρ_μ-dense in M^r by Lemma 3.28, it follows, by applying the triangle inequality, that $\Sigma\Pi^r(G)$ is ρ_μ-dense in M^r. □

4 Understanding Generative Networks

In this section, we will discuss about two classic types of the Generative Adversarial Network (GAN), which has worked as a catalyst for turning machine-learning and AI into the main stream in academia and industry today. As mentioned earlier, they are unsupervised learning algorithms that automatically tune the parameters of the DNNs typically included as its supervised learning modules.

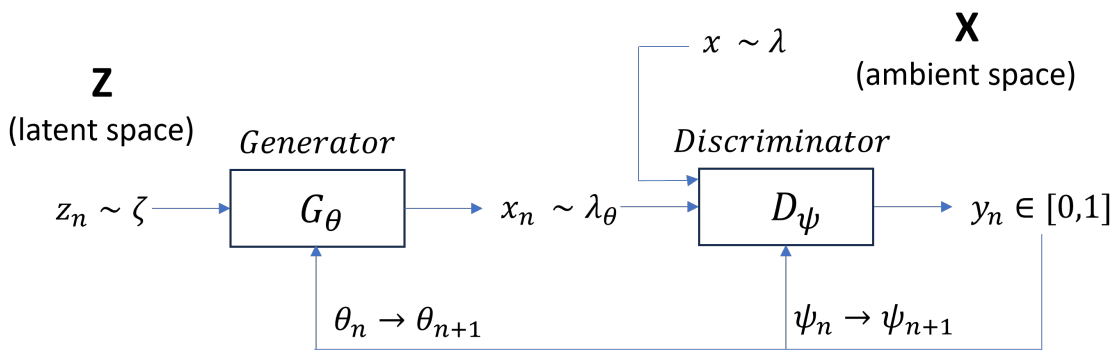


Figure 3: Generative Adversarial Network

The first one is f-GAN, which adopts f-divergence as a metric in tuning the parameters of the DNN, while the second one is W-GAN, which uses the Wasserstein norm as a metric for the same goal. We will be able to understand how the performance metric each of the two GANs is striving to optimize is derived from the perspective of measure theory. This can often help understand and improve numerous variants of these algorithms in a principled way. In addition, we can potentially come up with a new design through the knowledge and intuitions obtained during the process.

4.1 Structure of GAN

In machine learning, input data is said to belong to the *ambient space* while a condensed form of input data – called feature – is said to belong to the *latent space*. In fact, learning to well represent the essential information from the input data in the latent space is a key to machine learning algorithms. This is because differences and similarities inherent in input data can be revealed in a more lucid way such that tasks such as comparison, classification, and prediction may be performed more accurately in that space.

Generative Adversarial Networks (GANs) represent a non-cooperative game between two neural networks - the Generator and the Discriminator. They act as adversaries in the game with directly opposite goals, but end up improving each other through multiple cycles of competition.

Definition 4.1. Generator The Generator pulls samples (typically random noise) z from the latent space Z with probability measure ξ to generate the output $x_n = G(z;\theta_n)$ with the probability density $f_g(x_n;\theta_n)$ and the probability measure λ_n. It wants to make the output samples as close as the samples from real data in the ambient space with the probability density $f_{real}(x)$ and the probability measure λ.

Definition 4.2. Discriminator The Discriminator is given an input x_n and produces an output number $\mathrm{D}(x_n;\psi_n) \in [0,1]$, which is the probability that the input x_n is from the real data in the ambient space X with the probability measure λ.

Figure 3 shows how GAN works at each cycle n, where the Generator and the Discriminator generate outputs as explained in Definitions 4.1 and 4.2 above and then update their parameters (i.e., $\theta_n \to \theta_{n+1}$ and $\psi_n \to \psi_{n+1}$). In a nutshell, the Generator improves itself to fool the Discriminator by trying to make $\mathrm{D}(\mathrm{G}(z))$ close to 1 so that it believes the data generated by Generator is a real input data while the Discriminator improves itself to detect if the generated data is fake by trying to make $\mathrm{D}(\mathrm{G}(z))$ close to 0.

Now, let X and Z denote the ambient and latent spaces equipped with the measure λ and ξ, respectively. The GAN solves the following minimax game:

$$\min_{\mathrm{G}} \max_{\mathrm{D}} l_{\mathrm{GAN}}(\mathrm{D},\mathrm{G}) \tag{17}$$

where G represents the Generator which maps a latent vector z to a sample vector in the ambient space and D represents the Discriminator which takes the mapped sample vector as an input to differentiate it from the true sample in the ambient space. The quantity $l_{\mathrm{GAN}}(\mathrm{D},\mathrm{G})$, which G attempts to minimize while D attempts to maximize, represents the differentiation power and is defined as follows:

$$\begin{aligned} l_{\mathrm{GAN}}(\mathrm{D},\mathrm{G}) &\triangleq \mathrm{E}_{x\sim f_{real}}[\log \mathrm{D}(x)] + \mathrm{E}_{x\sim f_g}[\log(1-\mathrm{D}(x))] \\ &= \mathrm{E}_{\lambda}[\log \mathrm{D}(x)] + \mathrm{E}_{\xi}[\log(1-\mathrm{D}(\mathrm{G}(z)))] \\ &= \int_{\mathrm{X}} \log \mathrm{D}(x)\, d\lambda(x) + \int_{\mathrm{Z}} \log(1-\mathrm{D}(\mathrm{G}(z)))\, d\xi(z). \end{aligned} \tag{18}$$

Note that, for the Discriminator, maximizing the first term ensures a correct classification of the real data (with $\mathrm{D} \to 1$) while maximizing the second term ensures a correct classification of the generated data (with $\mathrm{D} \to 0$). Since the Generator's goal is exactly the opposite, it attempts to minimize the above quantity.

Multiple iterations of the following steps are performed until the generated data becomes indistinguishable from the real data:
Step 1. Given G, the optimal Discriminator $\mathrm{D}_{\mathrm{G}}^{opt}$ is obtained as

$$\mathrm{D}_{\mathrm{G}}^{opt} \triangleq arg \max_{\mathrm{D}} l_{\mathrm{GAN}}(\mathrm{D},\mathrm{G}).$$

Step 2. Given $\mathrm{D}_{\mathrm{G}}^{opt}$, the optimal Generator $\mathrm{G}_{\mathrm{D}}^{opt}$ is obtained as

$$\mathrm{G}_{\mathrm{D}}^{opt} \triangleq arg \min_{\mathrm{G}} l_{\mathrm{GAN}}(\mathrm{D}_{\mathrm{G}}^{opt},\mathrm{G}).$$

4.2 f-GAN and f-divergence [14]

The idea of f-GAN is to use f-divergence as a statistical distance between the real data distribution X with the measure λ and the synthesized data distribution with the measure μ so that the probability measure μ gets close to λ. We first introduce some concepts from measure theory in order to understand the f-divergence.

Definition 4.3. Absolutely continuous measure[4]
A measure λ on $\mathscr{X}$ is said to be absolutely continuous with respect to a measure μ on $\mathscr{X}$, or $\lambda << \mu$, if $E \in \mathscr{X}$ and $\mu(E) = 0$ imply that $\lambda(E) = 0$.

Definition 4.4. Radon-Nikodym Theorem [4]
Let λ and μ be a σ-finite measure defined on $\mathscr{X}$ and suppose that λ is absolutely continuous with respect to μ. Then there exists a function $f \in M^+(X, \mathscr{X})$ such that

$$\lambda(E) = \int_E d\lambda = \int_E f d\mu, \; E \in \mathscr{X}. \tag{19}$$

The function f is called the Radon-Nikodym derivative or density of λ with respect to μ and denoted as $d\lambda/d\mu$.

Definition 4.5. f-Divergence
Let λ and μ are two probability measures over a space $\mathscr{X}$ such that $\lambda << \mu$. Further assume they possess, respectively, the absolutely continuous density p and q with respect to the base measure x. In other words, $\lambda << x$ and $\mu << x$ with $d\lambda = pdx$ and $d\mu = qdx$ for the base measure x. Then for a convex, lower-semicontinuous function[20] f such that $f(1) = 0$, the f-divergence of μ from ν is defined as

$$\begin{aligned} D_f(\lambda||\mu) &\triangleq \int_{\mathscr{X}} f(\frac{d\lambda}{d\mu}) d\mu \\ &= \int_{\mathscr{X}} q(x) f(\frac{p(x)}{q(x)}) dx \end{aligned} \tag{20}$$

where $d\lambda/d\mu$ is the Radon-Nikodym derivative with respect to μ.

Definition 4.6. Every convex, lower-semicontinuous function [20] such that $f : T \to R$ has a **convex conjugate** function f^*, and it is defined as

$$f^*(x) = \sup_{\tau \in T} \{x\tau - f(\tau)\}.$$

Then f^* is again convex and lower-semicontinuous, and we have

$$f(x) = f^{**}(x) = \sup_{\tau \in T^*} \{x\tau - f^*(\tau)\},$$

when $f^* : T^* \to R$.

Proposition 4.7. *[14] Let $\lambda << \mu$. Then for an arbitrary function $\tau : X \to R$, we have a lower bound $D_f(\lambda||\mu) \geq \sup_{\tau\in T}\{\int_X \tau(x)d\lambda(x) - \int_X f^*(\tau(x))d\mu(x)\} = \sup_{\tau\in T}\{\mathbb{E}_{x\sim\lambda}[\tau(x)] - \mathbb{E}_{x\sim\mu}[f^*(\tau(x))]\}$, where $f^* : T \mapsto R$ is the convex conjugate of f and $\mathbb{E}$ denotes the expected value.*

Proof.

$$\begin{aligned}
D_f(\lambda||\mu) &= \int_X q(x) f(\frac{p(x)}{q(x)})dx \\
&= \int_X q(x) \sup_{\tau\in T}\{\tau\frac{p(x)}{q(x)} - f^*(\tau)\}dx \\
&\geq \sup_{\tau\in T}\int_X q(x)\{\tau\frac{p(x)}{q(x)} - f^*(\tau)\}dx \\
&= \sup_{\tau\in T}\int_X \tau(x)p(x)dx - \int_X q(x)f^*(\tau(x))dx \\
&= \sup_{\tau\in T}\int_X \tau(x)d\lambda(x) - \int_X f^*(\tau(x))d\mu(x) \\
&= \sup_{\tau\in T}\mathbb{E}_{x\sim\lambda}[\tau(x)] - \mathbb{E}_{x\sim\mu}[f^*(\tau(x))]
\end{aligned}$$

Here the second equality comes from Definition 4.6. Also, it can be shown[16] that the lower bound is tight and can be achieved with $\tau = f'(\frac{d\lambda}{d\mu}) = f'(\frac{p(x)}{q(x)})$ when $d\lambda = pdx$ and $d\mu = qdx$ for a base measure x. □

4.3 Target function for f-GAN[14]

Now we are going to show that by minimizing the f-divergence with certain design choices, we arrive at the target metric function which is exactly in the same form as the general GAN-metric as was given earlier by (18).

Definition 4.8. Push-Forward of a measure[19]

For $T : X \to Y$, the push-forward measure β of α by T, denoted as $T_{\#\alpha}$ with $\beta \in M(Y)$ and $\alpha \in M(X)$, satisfies

$$\int_Y h(y)d\beta(y) = \int_X h(T(x))d\alpha(x) \text{ , for } h \in C(Y).$$

Equivalently, for any measurable set $B \subset Y$, we have

$$\beta(B) = \alpha(\{x \in X : T(x) \in B\}) = \alpha(T^{-1}(B)). \tag{21}$$

Theorem 4.9. ***Change of variables formula*** *([3],Sec. 39)*

If f is a measurable function that maps a measure space (X, M, λ) into a measurable space (Y, N) such that a push-forward measure μ is defined by $\mu \triangleq f_{\#\lambda}$, and if g is an

extended real valued measurable function on Y*, then*

$$\int_{\mathrm{N}} g d\mu = \int_{\mathrm{M}} g \circ f d\lambda, \tag{22}$$

where $g \circ f$ *denotes the composite function.*

Remark 4.10. Note from Proposition 4.7 that we need to respect the domain of the conjugate function f^* when determining τ. For this, we can define $\tau(x) \triangleq g_f(\mathrm{V}(x))$, where $\mathrm{V} : \mathrm{X} \mapsto \mathrm{R}$ without any constraint on the output range while $g_f : \mathrm{R} \mapsto \mathrm{T}^*$ is an output activation function mapping the output to the domain of f^*.

Remark 4.11. Specifically for GAN, one can choose $f(x) \triangleq -(x+1)\log(x+1) + x\log x$, and obtain $f^*(u) = \sup_{x\in \mathrm{R}_+}\{ux + (x+1)\log(x+1) - x\log x\} = -\log(1-e^u)$ (see Table 1 and Table 2 of [14]). Therefore, the domain of f^* should be R_- in order to make $1 - e^u > 0$. Under this condition, we can choose $g_f(\mathrm{V}) \triangleq -\log(1+e^{-\mathrm{V}}) = \log(\mathrm{D})$ by defining $\mathrm{D} \triangleq \frac{1}{1+e^{-\mathrm{V}}}$, which will lead to $f^*(g_f(\mathrm{V})) = -\log(1-\mathrm{D})$.

With the design choices described in Remark 4.11, we now have

$$\begin{aligned}
\mathrm{D}_f(\lambda||\mu) &= \sup_{\tau\in \mathrm{T}^*} \int_{\mathrm{X}} \tau(x) d\lambda(x) - \int_{\mathrm{X}} f^*(\tau(x)) d\mu(x) \\
&= \sup_{g_f,\mathrm{V}} \int_{\mathrm{X}} g_f(\mathrm{V}(x)) d\lambda(x) - \int_{\mathrm{X}} f^*(g_f(\mathrm{V}(x))) d\mu(x) \\
&= \sup_{\mathrm{D}} \int_{\mathrm{X}} \log \mathrm{D}(x) d\lambda(x) + \int_{\mathrm{X}} \log(1-\mathrm{D}(x)) d\mu(x).
\end{aligned}$$

Observing that the measure μ is for the samples generated by generator $\mathrm{G}(z)$ with $z \in \mathrm{Z}$, where Z is the latent space with the measure ξ, we note μ is nothing but the push-forward measure $\mathrm{G}_{\#\xi}$. Using Theorem 4.9 (change-of-variables formula), the final loss function is thus given by

$$l(\mathrm{D},\mathrm{G}) \triangleq \sup_{\mathrm{D}} \int_{\mathrm{X}} \log \mathrm{D}(x) d\lambda(x) + \int_{\mathrm{Z}} \log(1-\mathrm{D}(\mathrm{G}(z))) d\xi(z). \tag{23}$$

Since Generator tries to minimize the above loss function, we finally get the quantity f-GAN is trying to optimize as follows:

$$\min_{\mathrm{G}} l(\mathrm{D},\mathrm{G}) = \min_{\mathrm{G}} \max_{\mathrm{D}} \int_{\mathrm{X}} \log \mathrm{D}(x) d\lambda(x) + \int_{\mathrm{Z}} \log(1-\mathrm{D}(\mathrm{G}(z))) d\xi(z). \tag{24}$$

4.4 W-GAN and Wasserstein metric[15]

The idea of W-GAN is to use the Wasserstein metric as a statistical distance between the real data distribution X with the measure λ and the synthesized data distribution with the measure μ so that the probability measure μ gets close to λ. Note that, unlike the f-divergence, the Wasserstein metric is one that satisfies all four properties of a metric.

Definition 4.12. Wasserstein Metric

Let (M, d) be a metric space with a metric d. For $p \geq 1$, let $P_p(M)$ denote the collection of all probability measures λ on M with a finite p-th moment. Then the p-th Wasserstein distance between two probability measures λ and μ in $P_p(M)$ is defined as

$$\begin{aligned} W_p(\lambda, \mu) &\triangleq (\inf_{\pi \in \Pi(\lambda,\mu)} \int_{M \times M} d(x, y)^p d\pi(x, y))^{1/p} \\ &= (\inf_{\pi \in \Pi(\lambda,\mu)} E_\pi[d(x, y)^p])^{1/p} \end{aligned} \tag{25}$$

where $\Pi(\lambda, \mu)$ denotes the collection of all measures on $M \times M$ with marginals λ and μ on the first and second factors, respectively, and X and Y are random vectors with the joint distribution π.

We say $T : X \mapsto Y$ transports a measure λ onto another measure μ, if

$$\mu(B) = \lambda(T^{-1}(B)) \tag{26}$$

for all μ-measurable sets B. In other words, μ is the push-forward of λ by T, or $\mu = T_{\#\lambda}$.

Definition 4.13. Optimal Transport

Suppose there are two spaces X and Y with measures λ and μ, respectively, linked through a map $T : X \to Y$. Furthermore, assume that there is a function $c : X \times Y \longmapsto R \cup \{\infty\}$ that represents the cost of moving one unit of mass from $x \in X$ to $y \in Y$. The optimal transport (OT) problem is to find a transport map T that transports λ to μ at the minimum total transportation cost as follows:

$$\min_T \{\int_X c(x, T(x)) d\lambda(x) : \mu = T_{\#\lambda}\} \tag{27}$$

The optimization problem (27) is difficult to solve in general due to the push-forward constraint. Kantorovich proposed to replace the deterministic transport map T by a probabilistic one as follows.

Definition 4.14. Kantorovich relaxation[19, 18] Let the joint measure $\pi \in P(X \times Y)$, where $P(X \times Y)$ is the set of probability distributions over product space $X \times Y$. Then the original optimal transport problem can be relaxed as

$$\min_\pi \int_{X \times Y} c(x, y) d\pi(x, y) \tag{28}$$

such that $\pi(A \times Y) = \lambda(A)$ and $\pi(X \times B) = \mu(B)$ for all measurable sets $A \in X$ and $B \in Y$.

Note that the conditions $\pi(A \times Y) = \lambda(A)$ and $\pi(X \times B) = \mu(B)$ can be equivalently written as $\int_Y d\pi(x, y) = d\lambda(x)$ and $\int_X d\pi(x, y) = d\mu(y)$, respectively. This simply means all the mass taken from point x sum up to $d\lambda(x)$ while all the mass transported to y coincide with $d\mu(y)$.

Theorem 4.15. ***Kantorovich Duality Theorem*** *[17, 18] Let* (X, λ) *and* (Y, μ) *be probability spaces and let* $c : X \times Y \longmapsto R$ *be a lower semicontinuous cost function such that* $c(x, y) \geq \varphi(x) + \psi(y)$ *for some* $\varphi(x) \in L^1(\lambda)$ *and* $\psi(y) \in L^1(\mu)$*, where* $L^1(\lambda)$ *and* $L^1(\mu)$ *denote the spaces of Lebesgue integrable functions[6] with respect to the measures* λ *and* μ*, respectively. Then there is duality:*

$$\begin{aligned}
&\min_{\pi \in \Pi(\lambda,\mu)} \int_{X \times Y} c(x, y) d\pi(x, y) \\
&= \sup_{\varphi \in L^1(\lambda)} \int_X \varphi(x) d\lambda(x) + \int_Y \varphi^c(y) d\mu(y) \\
&= \sup_{\psi \in L^1(\mu)} \int_X \psi^c(x) d\lambda(x) + \int_Y \psi(y) d\mu(y)
\end{aligned}$$

where $\Pi(\lambda, \mu) \triangleq \{\pi | \pi(A \times Y) = \lambda(A), \pi(X \times B) = \mu(B)\}$ *and* φ *and* ψ *are called the Kantorovich potentials with their c-transforms defined as:*

$$\begin{aligned}
\varphi^c &\triangleq \inf_x \{c(x, y) - \varphi(x)\} \\
\psi^c &\triangleq \inf_y \{c(x, y) - \psi(y)\}
\end{aligned}$$

4.5 Target function for W-GAN[15]

Similarly with f-GAN, we are going to show that by minimizing the Wasserstein metric with certain design choices, we arrive at the target metric function which is exactly in the same form as the general GAN-metric as was given earlier by (18). Specifically, W-GAN aims to minimize the following Wasserstein-1 norm:

$$W_1(P, Q) \triangleq \min_{\pi \in \Pi(\lambda,\mu)} \int_{X \times X} ||x - x'|| d\pi(x, x') \tag{29}$$

where X is the ambient space, λ and μ are measures for the real and generated samples x and x', respectively, and $\pi(x, x')$ is their joint distribution.

By applying Theorem 4.15 to the Wasserstein 1-norm above, we obtain

$$\begin{aligned}
W_1(\lambda, \mu) &= \sup_{\varphi \in L^1(\lambda)} \int_X \varphi(x) d\lambda(x) + \int_X \varphi^c(x') d\mu(x') \\
&= \sup_{\varphi \in Lip_1(X)} \{\int_X \varphi(x) d\lambda(x) - \int_X \varphi(x') d\mu(x')\},
\end{aligned} \tag{30}$$

where the last equality follows when we restrict φ to $Lip_1(X)$ which is the 1-Lipschitz function space with domain X satisfying the following condition:

$$|f(x) - f(y)| \leq ||x - y|| \text{ , for x,y } \in X.$$

Therefore, minimizing this quantity over the possible measure μ amounts to the following:

$$\min_{\mu} W_1(\lambda, \mu) = \min_{\mu} \max_{\varphi \in \mathrm{Lip}_1(X)} \{\int_X \varphi(x) d\lambda(x) - \int_X \varphi(x') d\mu(x')\} \tag{31}$$

Assuming G is the mapping used to generate the sample x' with μ as the push-forward measure of ξ from the latent space z, we get the following by the change-of-variable formula:

$$\begin{aligned} \min_{\mu} W_1(\lambda, \mu) &= \min_{G} W_1(\lambda, \mu) \\ &= \min_{G} \max_{\varphi \in \mathrm{Lip}_1(X)} \{\int_X \varphi(x) d\lambda(x) - \int_Z \varphi(G(z)) d\xi(z)\} \end{aligned} \tag{32}$$

where $G(z)$ is called the Generator and the Kantorovich potential φ is called the Discriminator.

5 Conclusion

In this paper, we provided a guided tour of mathematical aspects of some of the most fundamental and popular machine learning algorithms. The examples such as deep neural networks and generative adversarial networks discussed in this paper are considered at the core of ongoing machine learning research, which is driving the innovations in the area of AI.

The relevant tools and concepts from functional analysis and measure theory were also introduced and explained along the way and one can clearly appreciate the value and power of such mathematical tools and approaches in understanding the inner-workings of the basic building blocks of modern AI systems.

Furthermore, as in numerous traditional areas of applications in engineering and science where rigorous quantitative results are crucial, such a mathematical understanding will also provide useful insights that could facilitate developments of new ideas and solutions in the area of machine learning and AI.

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Brandon Yea
Henry M. Gunn High School, Palo Alto, CA 94306
brandon.nyea@gmail.com